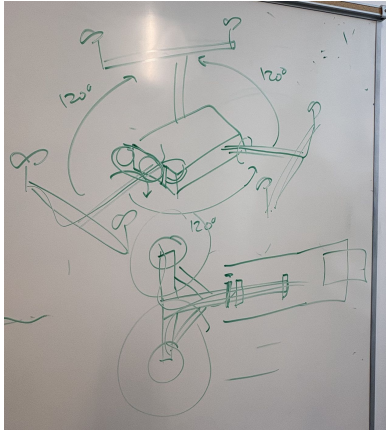


GeoDesiC: Hexrotor Drone

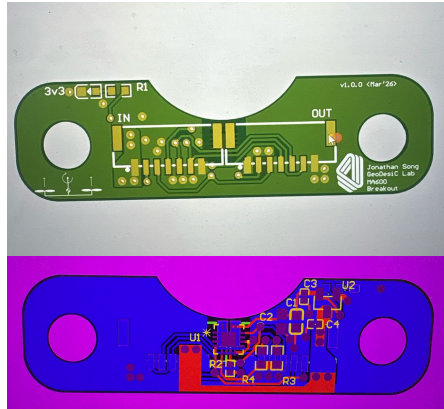
System Context:

- Tilt-rotor hexrotor with passively sensed dual-rotor joints
- Rotor thrust influences both joint angle and vehicle motion
- Uses magnetic encoders + custom electronics for joint state estimation
- Electrical role: sensor integration + signal routing + control interface

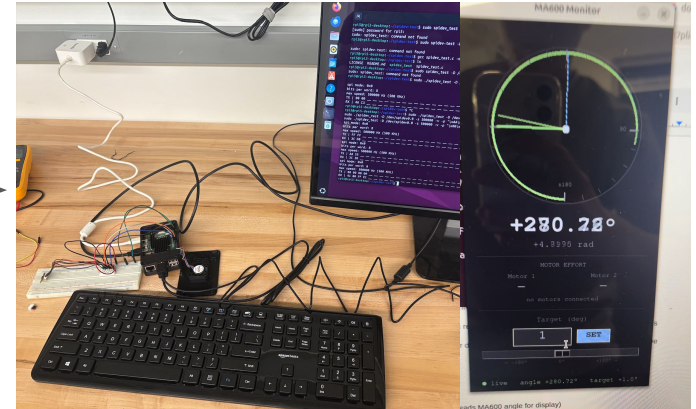
System Ideation



PCB Design



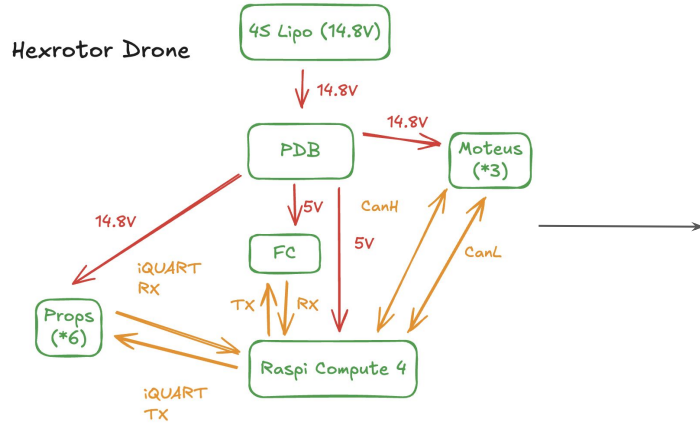
Sensor+Firmware Testing



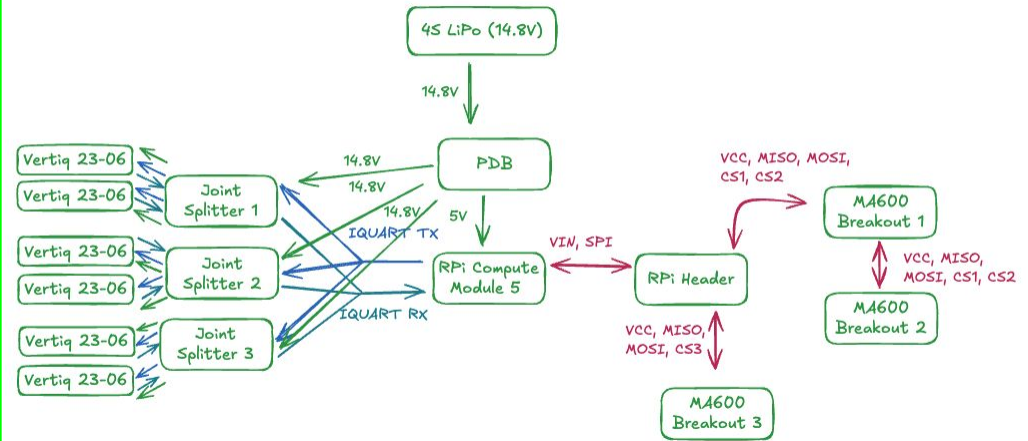
GeoDesiC: Hexrotor Drone

Electrical Architecture

Old

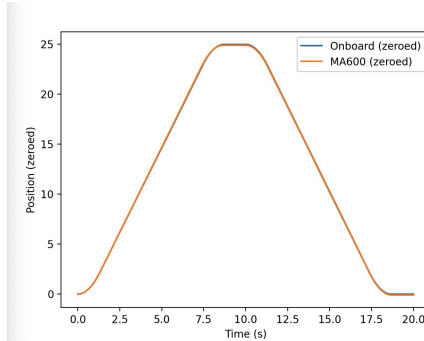
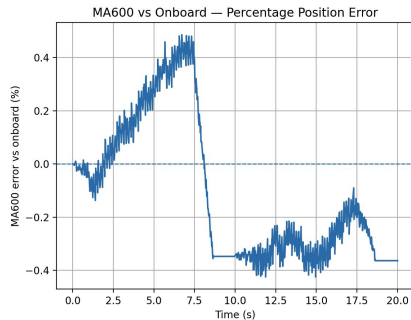
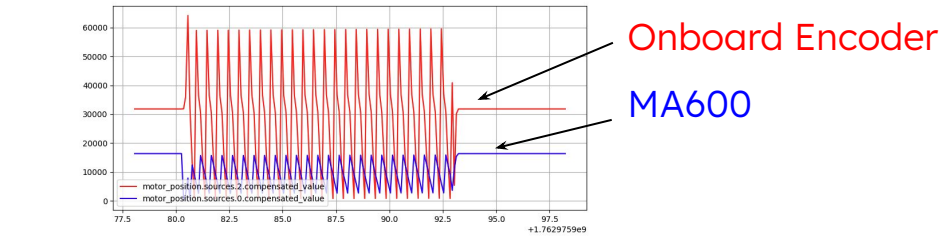


Current



GeoDesiC: Hexrotor Drone

Initial Encoder Testing using Moteus Controller Board

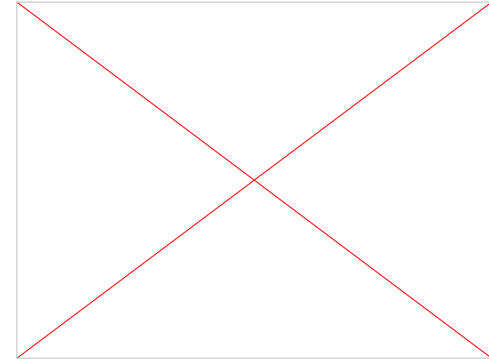
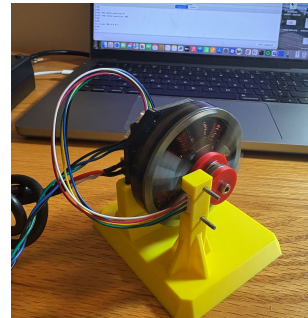


MA600 validated as viable joint sensing source

Compared against onboard encoder MA600 shows consistent behavior

MA600 Commutation Test

- Calibrated Motor Controller using MA600 as commutation source



GeoDesiC: Hexrotor Drone

PCB Integration

MA600 Breakout PCB

MA600 IC

Onboard
LDO

Daisy Chainable
Breakouts

Mechanical
Constraints

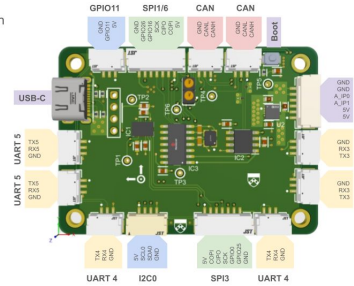
Joint Splitter PCB

Through holes for
Power, Signal Wires
(~24AWG)

M2.5 Mounting Holes

Pixhawk Standard/SPI

Integration
View



Existing Lab Pi Hat
(2 SPI Ports)

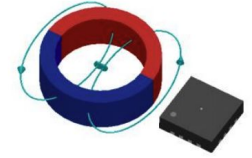


Figure 6: Side-Shift Mounting

Mounting Orientation

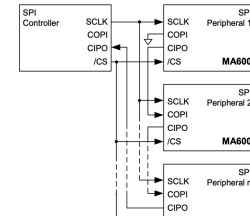
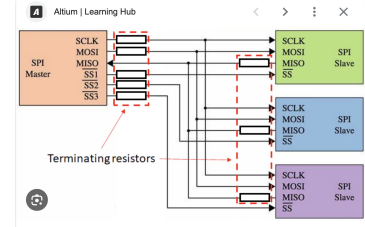


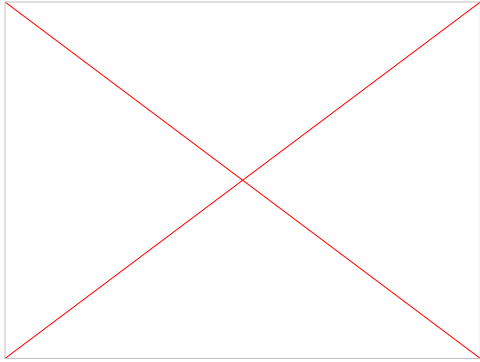
Figure 45: Daisy Chain Mode Configuration



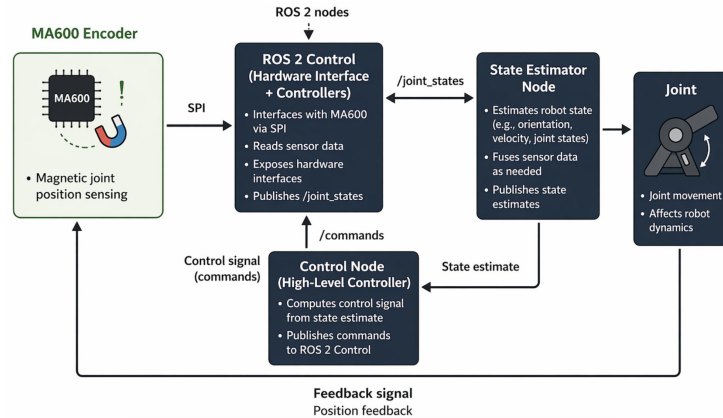
Tradeoff: "Fake" Daisy Chaining

GeoDesiC: Hexrotor Drone

Firmware/Debug

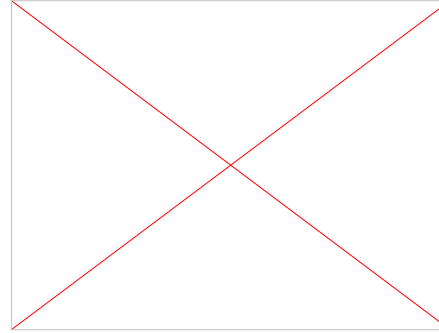


- Successful SPI readout of joint position
- Integrated with test setup + basic PID loop

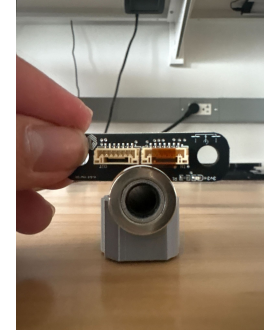
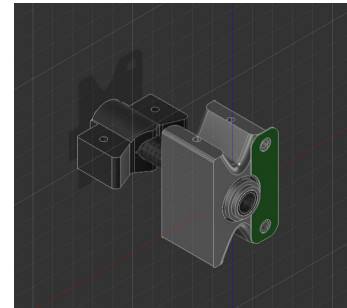


Sensor → Control → Actuation Loop

Results



Passive joint rotation



MA600 breakout
assembled + power
validated

PCB Integration
in Assembly